STUDENT ASSIGNMENT / FULL INSTRUCTIONS

Improve and Test the Agentic RAG Notebook

Start here: Colab notebook | Jev quick start

The notebook answers questions using OpenAI documentation, company financial reports, and live internet search. Its **router** decides which source to use.

Add support for questions with multiple parts, replace the existing router with Jev, and compare the two routers. **Submit your work as one video walkthrough.**

Part 1 - Break complex questions into smaller questions

Split a question with several parts into smaller questions, called **sub-queries**. Answer each one and combine the results.

Example: "What was Uber's revenue in 2021, and what are the newest LLMs?" becomes two questions: one about Uber's revenue and one about the newest LLMs.

Implement `agentic_rag_multi(user_query)` in the provided cell. It should:

- Use `sub_queries()` to split the original question.
- Route and answer each sub-query separately. Different sub-queries may need different sources.
- Combine the answers into one clear response that answers every part and keeps the source references.

A simple question should still use one route. Avoid an extra answer-combining call when there is only one answer. If the split result is unreadable or incorrectly formatted, treat the original input as one question instead of crashing.

Part 2 - Replace the existing router with Jev

Use TypeSafe's Jev to choose the source for each query or sub-query. Configure these choices:

Route	Source
`OPENAI_QUERY`	OpenAI documentation
`10K_DOCUMENT_QUERY`	Company financial reports
`INTERNET_QUERY`	Live internet search

Give Jev a clear description of each choice and connect its selected route to the appropriate notebook function. The existing retrieval and answer-generation steps still produce the answer.

Keep the original router available so you can test both versions.

Part 3 - Compare Jev with the original router

Build an evaluation: a set of test questions with clear rules for checking the results. Include questions for all three sources, simple and multi-part questions, questions that need different sources, unclear questions, and attempts to override the routing rules.

For each test, define the expected route or behavior and what a successful answer should contain **before scoring the results**.

Run both routers on the same test set. Use the same sub-queries, data sources, tools, answer-generation model, and scoring rules. Following the shared slides, compare:

Measure	What to check
Accuracy	Did it choose the correct source? Did the final answer answer the question correctly? Report these separately.
Safety	Did it follow the system's rules and handle attempts to override them appropriately?
Speed	How long did routing take? How long did the complete response take?
Cost	How much did routing and the complete response cost? Explain any estimates.

Present a side-by-side results table. Explain where each router worked well, where it failed, and which approach you would choose. **Jev does not have to perform better. Your conclusion should match your evidence.**

Optional bonus - Cache answers without breaking access rules

This bonus is optional and ungraded. A **semantic cache** reuses answers to similar questions. **Role-based access control (RBAC)** decides which sources a user may access.

An engineer must not receive a finance-only answer just because a finance analyst previously saved it in the cache.

Implement `secure_agentic_rag_cached(user_id, user_query)`:

- Reject unknown users before any model call, embedding, or cache lookup.
- Route the question and check the user's permission **before looking in the cache**.
- Separate cached answers by role, or attach permission information and check it before returning an answer.
- Return the stored answer on a cache hit. Otherwise, run the pipeline and save the answer when appropriate.
- Never cache a denial. Skip cache lookup and storage for time-sensitive questions.
- Return `answer`, `status` (`HIT`, `MISS`, or `DENIED`), and `role`. Pass the notebook's provided self-check.

Briefly explain your cache design and what happens if a user's role changes. Keep caching disabled for the main router comparison, or report results with and without caching separately.

What to submit

Submit **one video walkthrough** showing:

- Your notebook changes and a simple question and multi-part question running successfully.
- The generated sub-queries, selected sources, and final answers.
- Your test questions, scoring method, comparison results, conclusions, and limitations.
- If you attempted the bonus: a cache miss, a cache hit, an unauthorized request being denied, and the self-check passing.

Your notebook should run from start to finish. Show actual outputs clearly and explain your work in your own words.